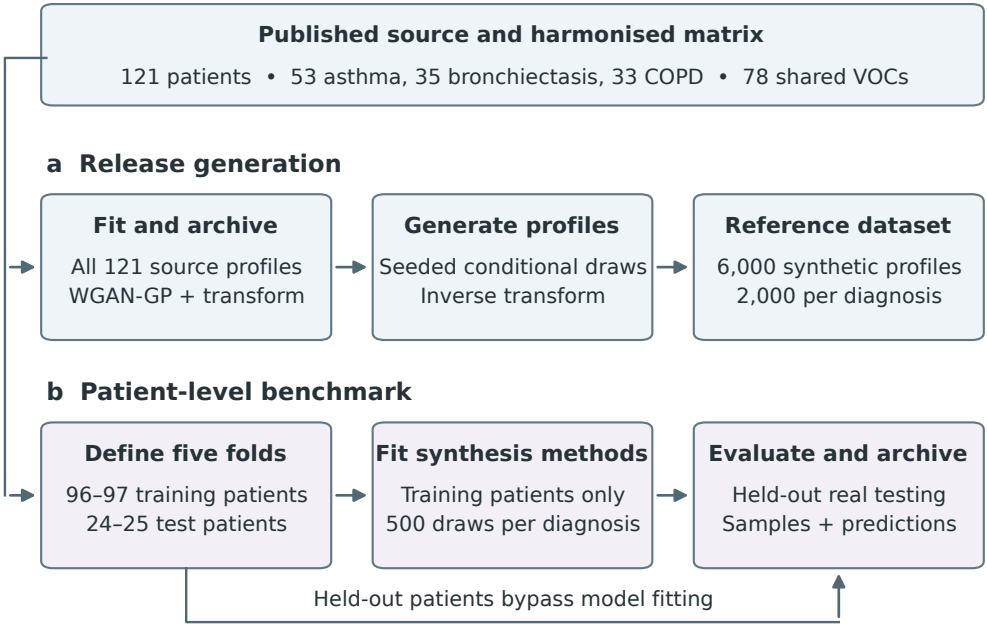
Published source and harmonised matrix
121 patients • 53 asthma, 35 bronchiectasis, 33 COPD • 78 shared VOCs
a Release generation
Fit and archive
All 121 source profiles
WGAN-GP + transform
Generate profiles
Seeded conditional draws
Inverse transform
Reference dataset
6,000 synthetic profiles
2,000 per diagnosis
b Patient-level benchmark
Define five folds
96–97 training patients
24–25 test patients
Fit synthesis methods
Training patients only
500 draws per diagnosis
Evaluate and archive
Held-out real testing
Samples + predictions
Held-out patients bypass model fitting